

Lorenzo Valente

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EDUCATION

By March 2027	PhD in Physics University of Hamburg, Germany Research Focus: Machine Learning for Fast Calorimeter Simulation, in collaboration with DESY. Supervisor: Prof. Gregor Kasieczka, co-supervisor Dr. Frank Gaede
March 2023	Master in Nuclear and Subnuclear Physics University of Bologna, Italy Thesis title: “A variational autoencoder application for real-time anomaly detection at CMS”. Supervisor: Prof. Daniele Bonacorsi
April – August 2021	Erasmus+ Exchange Semester Ludwig Maximilian University of Munich, Germany Courses: Machine Learning in Fundamental Physics (S. Krippendorf), Radiation Detectors for Medical Applications.
July 2020	Bachelor in Physics University of Bologna, Italy Thesis title: “Reconstruction of non-prompt charmed baryon Λ_c with Boosted Decision Trees technique” (ALICE experiment). Supervisor: Prof. Andrea Alici
October 2018	Master in Classical Piano Conservatory of Music Luisa D’Annunzio, Pescara, Italy

TALKS

Conference talks

○ European AI for Fundamental Physics Conference (EuCAIFCon)	Heidelberg, Germany
“Cross-Geometry Shower Simulation via Pre-training and Transfer”	24 August 2026
○ 14th Large Hadron Collider Physics Conference (LHCP)	Paris, France
“ML and Detector Simulation” (invited)	21 May 2026
○ 23rd International Workshop on Advanced Computing and Analysis Techniques in Physics Research (ACAT)	Hamburg, Germany
“Cross-Geometry Fast Electromagnetic Shower Simulation”	11 September 2025
○ 7th Inter-Experimental LHC Machine Learning Workshop	CERN, Switzerland
“Cross-Geometry Fast Electromagnetic Shower Simulation”	20 May 2025

Other talks

○ Glühwein Workshop 2025	Aachen, Germany
“Cross-Geometry Fast Electromagnetic Shower Simulation”	15 December 2025
○ 8th Round Table on Deep Learning at DESY	Hamburg, Germany
“Cross-Geometry Transfer Learning in Fast Electromagnetic Shower Simulation”	14 November 2025
○ DPG Spring Meeting 2025	Göttingen, Germany
“Adaptive Generative Modeling for Accelerated Calorimeter Simulations via Domain Transfer”	3 April 2025
○ DPG Spring Meeting 2024	Karlsruhe, Germany
“Adaptive Generative Modeling for High-Granularity Calorimeters”	8 March 2024
○ ML-INFN Weekly Meeting	INFN, Italy
“Joint Variational Auto-Encoder for Anomaly Detection in High Energy Physics”	17 July 2023

RESEARCH EXPERIENCE

- October 2023 – present **Doctoral Researcher**
University of Hamburg, Germany
Machine learning for fast calorimeter simulation, based on generative models and on transfer learning to unseen detector geometries.
Supervisors: Prof. Gregor Kasieczka and Dr. Frank Gaede, in collaboration with Dr. Katja Krüger
- April – July 2023 **Summer Student**
Instituto de Astrofísica de Canarias (IAC), Spain
Deep learning classification of heavy nuclei in cosmic rays (iron spectrum), ensemble approach reaching AUC 0.90.
Funded by an Erasmus+ traineeship grant.
Supervisors: Dr. Mónica Vázquez Acosta and Dr. Miguel Molero González
- June 2022 – March 2023 **Master's Thesis Research**
University of Bologna, Italy
Variational autoencoder for real-time anomaly detection at the CMS Level 1 trigger on public simulation, quantisation-aware training with QKeras and hls4ml.
Supervisor: Prof. Daniele Bonacorsi

TEACHING

- AI in Physics**, MSc course, University of Hamburg, tutor, 2025
- Python tutorial**, BSc course, University of Hamburg, tutor, 2025
- BSc thesis co-supervision**, Investigating the Behavior of Generative Models in Particle Physics under Recursive Training, University of Hamburg, 2025

SCHOOLS AND WORKSHOPS

- ELLIS Winter School on Foundation Models**, Amsterdam, Netherlands March 2025
- Generative Modeling Summer School**, Eindhoven, Netherlands July 2024
- Machine Learning with FPGAs for Particle Physics**, Bologna, Italy November 2022
- Quantum Sensing, Information Processing and Computing**, Bologna, Italy July 2022

TECHNICAL SKILLS

- Simulation & HEP Software:** ROOT, Geant4, DD4hep (ddsim, k4geo, ddfastsim), Key4hep, LCIO, uproot, awkward
- Programming Languages:** Python, C/C++, Bash, LabVIEW
- Machine Learning:** PyTorch, TensorFlow, Keras, scikit-learn, ONNX export, mixed-precision training, experiment tracking (Comet)
- Computing Infrastructure:** Slurm (DESY Maxwell), HTCondor (DESY NAF), Singularity containers, distributed multi-GPU training, HDF5
- Real-Time & FPGA:** hls4ml, QKeras, Vivado HLS, VHDL
- Software Engineering:** Git, CI, pytest, Docker

LANGUAGES

- ITALIAN (Native), ENGLISH (Fluent), SPANISH (B2), GERMAN (B1)

LORENZO VALENTE

PUBLICATIONS

PUBLICATIONS

* *corresponding author*

Journal articles and preprints

- F. Gaede, G. Kasieczka, **L. Valente***, *Cross-Geometry Transfer Learning in Fast Electromagnetic Shower Simulation*, JINST **21** (2026) P07037.
Contribution: conceived and implemented the transfer framework, ran all trainings and analysis, wrote the paper.
- H. Day-Hall*, ..., **L. Valente**, *CaloClouds3: Ultra-fast Geometry-Independent Highly-Granular Calorimeter Simulation*, JINST **21** (2026) P03018.
Contribution: trained and tuned successive model versions, fed improvements from my cross-geometry transfer work back into the model, contributed figures and manuscript review.
- T. Buss, ..., **L. Valente***, *Transferable Fast Calorimeter Shower Generation via Multi-Geometry Pre-training*, arXiv:2608.18233 (2026), alphabetical authors.
Contribution: led the study design, dataset production, training campaigns and analysis, wrote the paper.

Conference proceedings

- T. Buss, ..., **L. Valente***, *Point-Cloud Generative Models for Fast Calorimeter Simulation across Particles and Geometries*, arXiv:2609.30403 (2026), LHCP 2026, accepted in PoS.
Contribution: gave the invited talk and wrote the paper.
- T. Buss, ..., **L. Valente***, D. Weber, *Cross-Geometry Transfer and Model Collapse in Point Cloud Calorimeter Shower Generation*, arXiv:2609.28661 (2026), ACAT 2025, submitted to PoS.
Contribution: gave the talk and wrote the paper, the model collapse study comes from the BSc thesis I co-supervised.
- T. Buss, ..., **L. Valente**, *Jevons' Paradox and Fast Generative Simulation for HEP: Why Realistic Benchmarking is Essential*, arXiv:2609.24306 (2026), SciPost Phys. Proc. (submitted).
Contribution: co-author through CaloClouds3, the model the article builds on, which I trained and tuned across its versions.
- **L. Valente*** et al., *Joint Variational Auto-Encoder for Anomaly Detection in High Energy Physics*, PoS ISGC&HEPiX2023 (2023) 014.
Contribution: developed the model, ran the trainings and the analysis and wrote the paper, MSc work on real-time anomaly detection at CMS.